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TRENDS AND RANDOM WALKS IN MACROECONOMIC TIME SERIES
Further Evidence from a New Approach

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This paper presents a summary of recent work on a new methodology to test for the presence of a unit root in univariate time series models. The stochastic framework is quite general. While the Dickey–Fuller approach accounts for the autocorrelation of the first-differences of a series in a parametric fashion by estimating additional nuisance parameters, this new approach deals with this phenomenon in a nonparametric way. We apply these new tests to reassess recent findings on the behavior of common macroeconomic time series, including the various series studied by Nelson and Plosser (1982).

1. Introduction

In the time series representation of a variable, the hypothesis of a unit root has attracted an increasing amount of attention. As Blanchard (1981) pointed out, there seems to be an 'epidemic of martingales' in economics. The epidemic has taken both theoretical and empirical directions.

The presence of a unit root is often a theoretical implication of models which postulate the rational use of information available to economic agents. Examples from economics include various financial market variables such as future contracts [Samuelson (1965)], stock prices [Samuelson (1973)], dividends and earnings [Kleidon (1986)], spot and forward exchange rates [Meese and Singleton (1983)], and even aggregate variables like real consumption [Hall (1978)] and investment [Blanchard (1981)].

Tests of the unit root hypothesis are also of interest (even if it is not a theoretical implication of a formal model) because they help to evaluate the nature of the nonstationarity that most macroeconomic data exhibit. In

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particular, the presence of a unit root implies a stochastic nonstationarity instead of a deterministic one (such as a linear time trend). This distinction has profound implications for economic theory; in the former case, the random shocks have an enduring effect on future values of the variable, while in the latter, they have a vanishing effect. A recent examination of historical time series by Nelson and Plosser (1982), using the Dickey–Fuller (1979) testing procedure, found evidence consistent with a unit root nonstationarity in many macroeconomic time series.

The unit root hypothesis has additional theoretical and empirical implications. A good example can be found in the recent debate on variance bounds tests of the determination of stock prices as the discounted value (with a constant rate) of anticipated future dividends [see, for example, Shiller (1981a, 1984), LeRoy and Porter (1981), Marsh and Merton (1986), and Kleidon (1986)]. Tests for a unit root have also been advanced to detect the presence of 'bubbles' in models with rational expectations. For example, in Hamilton and Whiteman (1985, p. 366): 'if d-differences are sufficient to produce a stationary input series $(1-L)^d x_t$, then under the bubble hypothesis..., d-differences will not be sufficient to produce a stationary output series $(1-L)^d y_t$' [see also Diba and Grossman (1984)].

The extensive implications of the unit root hypothesis in the time series representation of a univariate model underscore the need to have reliable procedures to test formally this hypothesis. Investigations by Dickey (1976), Dickey and Fuller (1979, 1981), Fuller (1976), Evans and Savin (1981a, b, 1984) and Solo (1984) have been at the forefront of this research and provided a valuable set of tools to deal with this issue. However, the initial research was restricted to the case where the sequence of innovations driving the model are independent with common variance. Frequently, the assumption is that these innovations are i.i.d. $(0, \sigma^2)$ or that they are even i.i.d. $N(0, \sigma^2)$. Fuller (1976) and Dickey and Fuller (1979, 1981) extended their results to autoregressive models by considering regression equations with lags of first-differences of the data added as regressors.

Attention has also been given to more general ARIMA models. Said and Dickey's (1984) method is a natural extension of the one proposed by Dickey and Fuller. It deals with the problem of correlation in the first-differences of a variable by an autoregressive correction which adds extra lags of first-differences as regressors. The number of such extra lags needs to be an increasing function of the sample size T, at a controlled rate $T^{1/3}$. This approach, however, involves the estimation of additional nuisance parameters which reduces the effective number of observations due to the need for extra initial conditions.[1]

[1] For a review of the recent literature on this approach, see Fuller (1984) and Dickey, Bell and Miller (1986).

Recently, Phillips (1987a) has proposed an altogether different approach based on a nonparametric correction to account for the correlation in the first-differences of the variable. It has two important features. First, the procedure does not require the estimation of additional nuisance parameters with a reduction in the effective number of observations. Secondly, it is valid, in a much more general context, since weaker conditions are imposed on the stochastic innovations driving the system. This approach has been extended to allow for the presence of a nonzero mean and a nonzero drift, in the univariate case, by Phillips and Perron (1986) and Perron (1986b).

The purpose of this paper is to review the recent studies involving this new approach. The emphasis is practical with a focus on applications to aggregate macroeconomic variables. To this end, section 2 discusses the following topics: (a) the data generating mechanisms permitted under this new approach, (b) the various regression equations which can be used and their relation to the presence of nuisance parameters, (c) a presentation of Phillips' nonparametric approach and the new class of test statistics proposed, (d) the consistent estimation of variance estimators, (e) the statistical properties of the proposed statistics, and (f) a suggested testing strategy. Section 3 presents some empirical application of this new testing procedure. We first reassess the findings of Nelson and Plosser (1982) and conclude that their results are supported using these new statistics. We also examine the unit root hypothesis for a range of macroeconmic variables for which this hypothesis has been suggested, including consumption, dividends and earnings. Section 4 offers concluding comments.

2. Univariate time series models with a unit root

In this section, we present the possible data generating mechanisms under consideration and show the generality of the proposed approach. We also present a new set of statistics to test for the presence of a unit root which are valid under the stated conditions. Finally, we discuss the various options in the choice of the variance estimators and outline a testing strategy.

2.1. The data generating mechanisms

We consider two possible data generating mechanisms for the stochastic process $\{y_t\}_0^\infty$:

$$\begin{aligned} &(\mathrm{A}) \quad y_t = y_{t-1} + u_t, \\ &(\mathrm{B}) \quad y_t = \mu + y_{t-1} + u_t, \end{aligned} \qquad t = 1, 2, \ldots .$$

The first model specifies that $\{y_t\}_0^\infty$ is a driftless stochastic process with a unit root, while the second allows for the presence of a fixed drift. Both processes are nonstationary because of the presence of a unit root. Model (B) is likely to be the relevant one for most macroeconomic time series, for which we suspect the presence of a unit root; these series usually have a definite tendency to increase over time.

We allow the initial observation y_0 to be fixed at a given constant or to have a prespecified distribution.[2] Given this initial condition, we can write the process generating $\{y_t\}_0^\infty$ as

$$\text{(B}'\text{)} \qquad y_t = y_0 + \mu t + \sum_{j=1}^{t} u_j .$$

Therefore, y_t is a linear function of time [with a possible slope of zero when $\mu = 0$ as in model (A)], with intercept y_0 a function of historical events. The deviations from this function of time are nonstationary, as they are the accumulation of past random shocks. Indeed, the deviations from the linear function of time are nonstationary accumulations with a variance that increases over time. This implies that the variance of a forecast error on y_{t+k} increases without bound as k increases.

To complete the specification of the data generating mechanism, we need to impose some conditions on the innovation sequence $\{u_t\}_1^\infty$. These restrictions are closely linked to the asymptotic theory underlying the derivation of a nondegenerate limiting distribution. By inspection of (B′), with $\mu = 0$, the quantity of interest is the sum (or accumulation) of the random changes $S_t = \sum_{j=1}^{t} u_j$. What is needed are some conditions which would ensure that a suitably normalized function of S_t would have a nondegenerate asymptotic distribution.[3] A number of such sets of conditions are possible depending on the measure of temporal dependence used. The one adopted by Phillips

[2] We require, however, some regularity conditions on the distribution of y_0 such as the independence of its moments from the length of the sample size. Such a condition precludes some pathological cases where the initial observation has a distribution which becomes dominant as T, the sample size, tends to infinity. These cases would change the asymptotic theory underlying this method.

[3] More precisely, we need to transform the sequence $\{S_t\}_1^T$ to a random element which lies in the space $D(0,1)$ of all real valued functions on the interval $[0,1]$ that are right continuous and have finite left limits. This is achieved by defining the functions

$$X_T(r) = T^{-1/2}\sigma^{-1}S_{[Tr]} = T^{-1/2}\sigma^{-1}S_{j-1}, \quad (j-1)/T \le r < j/T, \quad j = 1,\dots,T,$$
$$X_T(1) = T^{-1/2}\sigma^{-1}S_T.$$

Upon the assumptions on the sequence $\{u_t\}_1^\infty$ given in the text we have $X_T(r) \to W(r)$; i.e., $X_T(r)$ converges weakly to a Weiner process (which lies in the space $C[0,1]$ of all real-valued functions continuous on the interval $[0,1]$).

(1987a), Phillips and Perron (1986) and Perron (1986b) is due to Herrndorf (1984a) and defines the measure of temporal dependence in terms of the mixing coefficient α_m defined as

$$\alpha_m = \sup_n \sup_{j>n+m} \alpha\left(F_1^n, F_{n+m}^j\right),$$

where

$$\alpha(F,G) = \sup_{f\in F, g\in G} |P(fg) - P(f)P(g)|,$$

and F_a^b denotes the σ-field generated by $\{u_a, u_{a+1}, \dots, u_b\}$ for all $a \le b$. A series is α-mixing (or strong mixing) if $\alpha_m \downarrow 0$ as $m \uparrow \infty$. The set of conditions on the innovation sequence $\{u_t\}$ can then be defined as follows:

Assumption 1

(a) $\mathrm{E}(u_t) = 0$ for all t.

(b) $\sup_t \mathrm{E}|u_t|^{\beta+\varepsilon} < \infty$ for some $\beta > 2$ and $\varepsilon > 0$.

(c) $\sigma^2 = \lim_{T\to\infty} T^{-1}\mathrm{E}(S_T^2)$ exists and $\sigma^2 > 0$.

(d) $\{u_t\}_1^\infty$ is strong mixing with mixing numbers α_m that satisfy

$$\sum_1^\infty \alpha_m^{1-2/\beta} < \infty.$$

The conditions[4] imposed on the innovation sequence $\{u_t\}_1^\infty$ by this set of assumptions are not overly restrictive and permit many weakly dependent and heterogeneously distributed time series. Condition (b) controls the allowable heterogeneity of the errors and (d) the extent of permissable temporal dependence in relation to the probability of outlier occurences. The presence of the coefficient β in conditions (b) and (d) suggests that there is a tradeoff between a higher probability of outliers and the extent of temporal dependence in the series. Condition (c) controls the normalization at a rate which will ensure a nondegenerate limit distribution. Note in particular, that $\{u_t\}_1^\infty$ need not be stationary. However, if it is stationary, then condition (c) is automatically satisfied and

$$\sigma^2 = \mathrm{E}\left(u_1^2\right) + 2\sum_{k=2}^\infty \mathrm{E}(u_1 u_k).$$

[4] Other sets of conditions are of course possible and would also ensure the relevant convergence result. They differ basically in the measure of temporal dependence adopted. For example, McLeish's (1975) weak convergence theorem uses α- and ρ-mixing measures of weak dependence; while Herrndorf (1984b) uses the maximal correlation coefficient ρ_m and the concept of ρ-mixing sequences. For a discussion of these issues, see Phillips (1987b).

The generality of the conditions imposed on the sequence $\{u_t\}_1^\infty$ implies that a wide variety of possible data generating mechanisms are permitted. These include virtually any ARMA model with a unit root and even ARMAX systems with a unit root and with stable exogeneous processes that admit a Wold decomposition. In the former case we have

$$a(L)(1-L)y_t = b(L)e_t,$$

which is in the form of model (A) with $u_t = a(L)^{-1}b(L)e_t$. In this case, $\{u_t\}_1^\infty$ will satisfy the set of assumptions above under very general conditions on the innovations $\{e_t\}$ and lag polynomials $a(L)$ and $b(L)$ [see Withers (1981)]. For example, they are satisfied if $\{e_t\}$ is Gaussian and the lag polynomials are finite. In the case of ARMAX systems, we can write

$$a(L)(1-L)y_t = b(L)x_t + c(L)e_t,$$

with

$$d(L)x_t = f(L)v_t,$$

which is in the form of model (A) with $u_t = a(L)^{-1}c(L)e_t + a(L)^{-1} \times b(L)d(L)^{-1}f(L)v_t$. Again, the sequence $\{u_t\}$ will satisfy the required assumptions upon general conditions on $\{e_t\}$, $\{v_t\}$ and the lag polynomials.

The central advantage of the new approach is that even if we permit a wide class of possible data generating mechanisms, the test statistics to be presented below require only the estimation of a first-order autoregression by least squares and a correction factor based on the structure of the residuals from this regression. This procedure is easy to implement.

2.2. *The regression equations*

We define three regression equations, all estimated by ordinary least squares, and outline their role in testing the null hypothesis that a sample of size $T+1$ of a univariate series $\{y_t\}$ is generated by either model (A) or (B),

$$y_t = \hat{\alpha} y_{t-1} + \hat{u}_t, \tag{1}$$

$$y_t = \mu^* + \alpha^* y_{t-1} + u_t^*, \tag{2}$$

$$y_t = \tilde{\mu} + \tilde{\beta}(t - T/2) + \tilde{\alpha} y_{t-1} + \tilde{u}_t. \tag{3}$$

Let $t_{\hat{\alpha}}$, t_{α^*} and $t_{\tilde{\alpha}}$ be the usual regression t-statistics for testing the null hypothesis that $\alpha = 1$ in eqs. (1), (2) and (3), respectively. Similarly, let t_{μ^*} ($\mu = 0$), $t_{\tilde{\mu}}$ ($\mu = 0$) and $t_{\tilde{\beta}}$ ($\beta = 0$) be the t-statistics on μ^*, $\tilde{\mu}$ and $\tilde{\beta}$ in their corresponding regressions.

For tests of joint hypotheses, we shall use the regression 'F-tests' defined by[5]

$$\Phi_1 = (2S^{*2})^{-1}\left[TS_0^2 - TS^{*2}\right]$$

$$\text{for } \mathrm{H}_0^1: \quad (\mu, \alpha) = (0,1) \text{ in } (2),$$

$$\Phi_2 = (3\tilde{S}^2)^{-1}\left[TS_0^2 - T\tilde{S}^2\right]$$

$$\text{for } \mathrm{H}_0^2: \quad (\mu, \beta, \alpha) = (0,0,1) \text{ in } (3),$$

$$\Phi_3 = (2\tilde{S}^2)^{-1}\left[T\left\{S_0^2 - (\bar{Y} - \bar{Y}_{-1})^2\right\} - T\tilde{S}^2\right]$$

$$\text{for } \mathrm{H}_0^3: \quad (\mu, \beta, \alpha) = (\mu,0,1) \text{ in } (3),$$

where $S_0^2 = T^{-1}\Sigma_1^T(y_t - y_{t-1})^2$, and S^{*2} and $\tilde{S}^2$ are the sample variances of the estimated residuals from regressions (2) and (3), respectively.

Regression equation (1), which contains neither a constant nor a trend as regressors, is appropriate in the driftless case [model (A)] when the initial observation y_0 is equal to 0.[6] Although the effects of a nonzero value of y_0 vanish asymptotically, they are substantial in finite samples as demonstrated by Evans and Savin (1981a) (for $\hat{\alpha}$) and Nankervis and Savin (1985) (for $t_{\hat{\alpha}}$),[7] in the case of i.i.d. errors. Under this assumption, the actual size of the test is well below the (asymptotic) nominal size. If the mean of the series were known, it could be subtracted from each observation and one could then use

[5] The finite sample expressions of the 'F-tests' are $(T-2)\Phi_1/T$, $(T-3)\Phi_2/T$ and $(T-2)\Phi_3/T$ [see Dickey and Fuller (1981)]. These latter expressions are the ones used in the empirical applications.

[6] In the model without drift, the mean of the series is determined by the initial observation under the hypothesis of a unit root. So the regression equation (1) is valid, in other words, when the overall mean of the series is 0.

[7] The relevant variable affecting the shape of the distribution in finite samples is y_0/σ in the case of i.i.d. $(0, \sigma^2)$ errors. For a Monte Carlo study of the effects of a nonzero value of y_0 with non i.i.d. errors, see Perron (1986a, ch. 4). An interesting point was made by Phillips (1987a) who showed that the asymptotic distribution of $T(\hat{\alpha}-1)$ was heavily influenced by y_0/σ, when T increases to infinity keeping a fixed span of data (i.e., continuous records asymptotic). This, in part, explains how y_0 affects the distribution of $T(\hat{\alpha}-1)$ and $t_{\hat{\alpha}}$ in a finite sample.

regression equation (1) as the basis for a unit root test. In practice, the mean is not likely to be known, so the presence of a nonzero mean for the series calls for regression model (2).

Regression equation (2), which incorporates a constant as a regressor, allows for a nonzero mean in the series. α^* and t_{α^*} are invariant with respect to y_0. However, their distribution is invariant only if we make the assumption that y_0 is fixed [see Evans and Savin (1984)]. The effects of y_0, however, are not substantial unless y_0 has a special distribution.[8] Of course, μ^* is not invariant with respect to y_0 but the joint test statistic Φ_1 is invariant. Therefore, regression equation (2) is acceptable to test for a unit root in the driftless case. If the drift is nonzero, then, even asymptotically, the distributions of the test statistics are affected.[9] So regression model (2) is not acceptable in this case.

Dickey (1976) showed that $\tilde{\alpha}$ and $t_{\tilde{\alpha}}$ are invariant with respect to μ, the drift term, if the data generating mechanism is (B). Regression equation (3) should be adopted if a nonzero drift is suspected as would be the case in most aggregate macroeconomic variables. Indeed, all statistics related to this regression model, except $\tilde{\mu}$ and $t_{\tilde{\mu}}$ of course, are invariant with respect to y_0 and $\tilde{\alpha}$, $t_{\tilde{\alpha}}$ and Φ_3 are invariant with respect to μ. (Note that $t_{\tilde{\beta}}$ is not invariant with respect to μ.) These statistics are, however, not invariant with respect to β, the trend parameter, even asymptotically.[10] In practice, this is not a problem since one usually wants to test for a series with a unit root and a drift *against* the hypothesis that the series is stationary around a linear trend. The hypothesis of a unit root with a trend is usually precluded a priori. For example, if the series is in logarithmic form, it implies an ever increasing (or decreasing) rate of change. [See Nelson and Plosser (1982).]

However, the most important noninvariance shared by all three regression models is that the limiting (as well as finite sample) distributions of all the statistics considered depend upon the correlation structure of the residuals. More precisely, the limiting distributions depend upon the ratio σ^2/σ_u^2, where $\sigma^2 = \lim_{T\to\infty} T^{-1}\mathrm{E}(S_T^2)$ ($S_T = \sum_{j=1}^{T} u_j$) and $\sigma_u^2 = \lim_{T\to\infty} T^{-1}\sum_1^T \mathrm{E}(u_t^2)$. When the errors are such that $\sigma^2 = \sigma_u^2$, the limiting distributions are invariant with respect to any nuisance parameters. The percentage points of these limiting distributions can, therefore, be obtained. For the statistic $T(\hat{\alpha}-1)$, these were found by Evans and Savin (1981b) by numerical integration. The percentage

[8]The limiting distribution of $T(\alpha^*-1)$ and t_{α^*} are the same as $T\uparrow\infty$ with the sampling interval fixed as it is when $T\uparrow\infty$ with the span fixed. [See Perron (1986a, ch. 2).]

[9]When $\mu \neq 0$, $T(\hat{\alpha}-1) \to 3/2$, $T^{1/2}(T(\hat{\alpha}-1)-3/2) \to \mathrm{N}(0,(9/5)\sigma^2/\mu^2)$, $T^{-1/2}t_{\hat{\alpha}} \to (\sqrt{3}/2)\mu(\sigma^2+\mu^2/4)^{-1/2}$, $T^{3/2}(\alpha^*-1) \to \mathrm{N}(0,12\sigma^2/\mu^2)$ and $t_{\alpha^*} \to \mathrm{N}(0,\sigma^2/\sigma_u^2)$, where $\sigma^2 = \lim_{T\to\infty} T^{-1}\mathrm{E}(S_T^2)$ and $\sigma_u^2 = \lim_{T\to\infty} T^{-1}\Sigma_1^T \mathrm{E}(u_t^2)$.

[10]In this case, we have the following asymptotic distributions [see Perron (1986a, ch. 2)]: $T^{1/2}(\tilde{\mu}-\mu) \to \mathrm{N}(0,6\sigma^2)$, $T^{3/2}(\tilde{\beta}-\beta) \to \mathrm{N}(0,12\sigma^2)$, $T^{5/2}(\tilde{\alpha}-1) \to \mathrm{N}(0,720\sigma^2/\beta^2)$, and the t-statistics all converge to $\mathrm{N}(0,\sigma^2/\sigma_u^2)$.

points of the limiting distribution of the other statistics [as well as $T(\hat{\alpha}-1)$] were obtained by Dickey and Fuller using a simulation method.[11]

To derive these asymptotic distributions, it was assumed that the errors were i.i.d. $(0, \sigma^2)$. It is easy to verify that this is a sufficient condition for $\sigma^2 = \sigma_u^2$; it is not, however, necessary. This equivalence also holds, for example, with innovations that are martingale differences under mild additional conditions [such as the moment condition (b) of Assumption 1]. Therefore, the limiting distributions obtained by Dickey and Fuller are also valid in the presence of some heterogeneity in the innovation sequence provided the innovations are martingale differences. In general, however, they cease to be appropriate when the innovations are nonorthogonal and $\sigma^2 \neq \sigma_u^2$.

When the innovation sequence $\{u_t\}_1^\infty$ is correlated, there is a need to either change the estimation method (adopt another regression model) or modify the statistics described above. Dickey and Fuller use the first approach. Consider first the case in which the series of first-differences $\{\Delta y_t\}$ has a stationary AR(p) representation with p known; i.e., $(1-L)a(L)y_t = e_t$ with $a(L)$ a pth-order polynomial in L.[12] Then, we can test the null hypothesis of a unit root by estimating an autoregression of Δy_t on its lags and y_{t-1} using OLS (here $l = p$):

$$\Delta y_t = \mu + \alpha y_{t-1} + \sum_{j=1}^{l} d_j \Delta y_{t-j} + \varepsilon_t. \tag{4}$$

If there is a unit root, $\alpha = 0$ and the t-statistic for α has the same limiting distribution as when the errors are i.i.d. and regression model (2) is used. The same principle holds for the t- and Φ-statistics derived from the other regression models. Note, however, that we can no longer use $T(\alpha^* - 1)$ [nor $T(\hat{\alpha}-1)$ and $T(\tilde{\alpha}-1)$] as the basis of a test statistic since they are not invariant to the true population value of the parameters $\{d_j, j = 1, \dots, p\}$.[13]

This method was recently extended by Said and Dickey (1984) to the more general case in which, under the null hypothesis, the series of first-differences are of the general ARMA (p, q) form with p and q unknown. They showed that a regression model, such as (4), is still valid, if the number of lags, l, of Δy_t introduced as regressors increases with the sample size at a controlled rate ($T^{1/3}$). In this case, the limiting distribution of the t-statistics of the coefficient on the lagged dependent variable y_{t-1} has the same noncentral limiting

[11] See Dickey (1976) [with tabulated values reproduced in Fuller (1976)] and Dickey and Fuller (1981). The basic method is to get a representation of the limiting distribution in terms of functions of infinite sums of normal random variables. For simulation purposes, the infinite sums are truncated in such a way that the sampling moments of each component correspond to their theoretical moments. See Dickey (1976) for a description of the methodology.

[12] In which case $u_t = a(L)^{-1} e_t$ is an infinite moving average process.

[13] A transformation of these statistics is, however, possible since the d's are consistently estimated in regressions such as (4). See Dickey and Fuller (1981).

distribution as discussed above in the case of the regression model (2) with i.i.d. errors.[14]

If the moving average terms are important, the number of extra lags of Δy_t, needed as regressors in the autoregressive correction, may be quite large. Furthermore, as shown by Schwert (1987), using Monte Carlo simulations, the exact size of the test may be far from the nominal size if the order of the autoregressive correction is not increased as the sample size increases. An adjustment is essential because the effects of the correlated structure of the residuals on the shape of the distribution becomes more precise. This point highlights the fact that with important MA components in the structure of the series $\{y_t\}$ a large number of nuisance parameters may be needed in the estimation. Since we furthermore lose one effective observation for each extra lag of Δy_t introduced, this implies that the Said and Dickey approach may have substantially lower power when MA terms are more important than if the errors were, say, i.i.d..

It is generally believed that moving average terms are present in many macroeconomic time series after first-differencing. Indeed, Nelson and Plosser (1982) argue that the majority of the macroeconomic variables they studied were adequately represented by an IMA(1) process.[15] Furthermore, there are several theoretical reasons which suggest the presence of a moving average structure. Examples are: time averaged data [e.g., Working (1964)], forward contracts when sampling interval is finer than the maturity of the contract [e.g., Meese and Singleton (1983)], an index of stock prices with infrequent trading for a subset of the index [e.g., Praetz (1982)], the presence of errors in the data, etc. In this light, it appears that the problems noted above with the Said and Dickey approach may have some practical consequences.

Accordingly, it would be desirable to have an approach for a test of the hypothesis of a unit root which takes into consideration the correlation structure of the residuals $\{u_t\}$ in a nonparametric way. Such an approach was first developed by Phillips (1987a) and subsequently extended by Phillips and Perron (1986) and Perron (1986b).[16]

2.3. Phillips' nonparametric approach

The nonparametric approach is valid under the conditions of Assumption 1 on the innovation sequence $\{u_t\}$. The idea is to use the theory of weak convergence to functionals of Weiner processes to characterize the limiting

[14] Note that in this case, no transformation of the normalized estimator $T(\alpha^* - 1)$ is possible since, theoretically, an infinite number of nuisance parameters affect the shape of the distribution, even asymptotically.

[15] The parameter of the MA(1) component is generally positive.

[16] See also Perron (1986a) for a detailed treatment of the univariate case. Multivariate extensions were considered in Phillips and Durlauf (1986).

distribution of the statistics. In general, when $\sigma^2 \neq \sigma_u^2$, these asymptotic distributions depend upon the nuisance parameter σ^2/σ_u^2. The larger the temporal dependence in the innovation sequence $\{u_t\}$, the greater will σ^2/σ_u^2 diverge from unity and the greater will be the effects on the shape of the limiting distribution compared to the reference case of i.i.d. errors.

The derivation of the limiting distribution of the statistics highlights the way in which the ratio σ^2/σ_u^2 affects the shape of the distribution. It is then possible to find a transformation of the various statistics which will eliminate the dependence of the limiting distribution on the nuisance parameters σ^2 and σ_u^2. Furthermore, the elimination of the dependence of the limiting distributions on σ^2/σ_u^2 can be accomplished in such a way that the transformed statistics converge, in the limit, to the same random variable (under general error structures) as do the untransformed statistics when the errors are i.i.d.. This implies that the critical values of the transformed statistics are the same as those tabulated by Dickey and Fuller.

Table 1 presents a summary of the test statistics proposed by Phillips (1987a) (a,b), Phillips and Perron (1986) (c,d,e,g,h,i,j) and Perron (1986b) (f,k,l). These statistics are transformations of the original regression statistics discussed above and analyzed by Dickey and Fuller. They also involve the estimators S_u^2 and S_{Tl}^2, which are consistent estimators of σ_u^2 and σ^2, respectively.[17]

As stated above, the purpose of the transformations is to obtain a test statistic whose limiting distribution is independent of the nuisance parameters σ_u^2 and σ^2. In effect, the proposed statistics change the original regression statistics by using sample information on the values of σ_u^2 and σ^2 to account for additional correlation in the innovations.

The transformations affect the limiting distribution of the original statistics generally both in its shape and its location. They affect the shape of the distribution via a scale parameter S_u/S_{Tl} which converges in the limit to σ_u/σ. Therefore, the shape is more concentrated as $\sigma_u^2 < \sigma^2$ (positive autocorrelation of the residuals) and vice versa. As well, the transformations affect the location via a factor $(S_{Tl}^2 - S_u^2)$ appropriately normalized by the relevant moment expression of the original data. Note that the shift in location is of opposite direction for negative and positive autocorrelation in the residuals.

These statistics, therefore, provide a relatively easy way to implement tests of hypotheses in time series regression with a unit root with possibly heterogeneously and dependently distributed data. Note again that the asymptotic critical values of the tests are the same as the asymptotic critical values tabulated by Dickey and Fuller. However, the finite sample counterparts are not the same and should not be used with the test statistics proposed here.

[17] The choices available to obtain consistent estimators of σ^2 and σ_u^2 are discussed in the next section.

Table 1

Summary of the test statistics.

Null hypothesis tested	Test statistics for a unit root in univariate models	Source for critical values
(a) $\alpha = 1$ in (1) ($y_0 = 0$)	$Z(\hat{\alpha}) = T(\hat{\alpha} - 1) - \frac{1}{2}\left(S_{Tl}^2 - S_u^2\right)\Big/\left(T^{-2}\sum y_{t-1}^2\right)$	Fuller (1976, p. 371)
(b) $\alpha = 1$ in (1) ($y_0 = 0$)	$Z(t_{\hat{\alpha}}) = (S_u/S_{Tl})t_{\hat{\alpha}} - \frac{1}{2}\left(S_{Tl}^2 - S_u^2\right)\left[S_{Tl}\left(T^{-2}\sum y_{t-1}^2\right)^{1/2}\right]^{-1}$	Fuller (1976, p. 373)
(c) $\alpha = 1$ in (2)	$Z(\alpha^*) = T(\alpha^* - 1) - \frac{1}{2}\left(S_{Tl}^2 - S_u^2\right)\left[T^{-2}\sum\left(y_{t-1} - \bar{Y}_{-1}\right)^2\right]^{-1}$	Fuller (1976, p. 371)
(d) $\alpha = 1$ in (2)	$Z(t_{\alpha}^*) = (S_u/S_{Tl})t_{\alpha^*} - (1/2S_{Tl})\left(S_{Tl}^2 - S_u^2\right)\left[T^{-2}\sum\left(y_{t-1} - \bar{Y}_{-1}\right)^2\right]^{-1/2}$	Fuller (1976, p. 373)
(e) $\mu = 0$ in (2)	$Z(t_{\mu^*}) = (S_u/S_{Tl})t_{\mu^*} + (1/2S_{Tl})\left(S_{Tl}^2 - S_u^2\right)\left[T^{-2}\sum\left(y_{t-1} - \bar{Y}_{-1}\right)^2\right]^{-1/2}$ $\cdot\left[T^{-2}\sum y_{t-1}^2\right]^{-1/2}\left(T^{-3/2}\sum y_{t-1}\right)$	Dickey–Fuller (1981, p. 1062)
(f) $\mu = 0$, $\alpha = 1$ in (2)	$Z(\Phi_1) = \left(S_u^2/S_{Tl}^2\right)\Phi_1 - \left(1/2S_{Tl}^2\right)\left(S_{Tl}^2 - S_u^2\right)$ $\cdot\left\{T(\alpha^* - 1) - \frac{1}{4}\left(S_{Tl}^2 - S_u^2\right)\left[T^{-2}\sum\left(y_{t-1} - \bar{Y}_{-1}\right)^2\right]^{-1}\right\}$	Dickey–Fuller (1981, p. 1063)
(g) $\alpha = 1$ in (3)	$Z(\tilde{\alpha}) = T(\tilde{\alpha} - 1) - (T^6/24D_x)\left(S_{Tl}^2 - S_u^2\right)$	Fuller (1976, p. 371)
(h) $\alpha = 1$ in (3)	$Z(t_{\tilde{\alpha}}) = (S_u/S_{Tl})t_{\tilde{\alpha}} - \left(T^3/4\sqrt{3}\,D_x^{1/2}S_{Tl}\right)\left(S_{Tl}^2 - S_u^2\right)$	Fuller (1976, p. 373)
(i) $\mu = 0$ in (3)	$Z(t_{\tilde{\mu}}) = (S_u/S_{Tl})t_{\tilde{\mu}} + \left(T^3/24D_x^{1/2}E_xS_{Tl}\right)\left(S_{Tl}^2 - S_u^2\right)\left(T^{-3/2}\sum y_{t-1}\right)$	Dickey–Fuller (1981, p. 1062)

(j) $\beta = 0$ in (3)	$Z(t_{\tilde{\beta}}) = (S_u/S_{Tl})t_{\tilde{\beta}} - \left(T^3/2D_x^{1/2}S_{Tl}\right)\left\{T^{-2}\Sigma\left(y_{t-1} - \overline{Y}_{-1}\right)^2\right\}^{-1/2} \cdot \left(S_{Tl}^2 - S_u^2\right)\left\{\frac{1}{2}T^{-3/2}\Sigma y_{t-1} - T^{-5/2}\Sigma(t-1)y_{t-1}\right\}$	Dickey–Fuller (1981, p. 1062)
(k) $\mu = 0$, $\beta = 0$, $\alpha = 1$ in (3)	$Z(\Phi_2) = \left(S_u^2/S_{Tl}^2\right)\Phi_2 - \left(1/3S_{Tl}^2\right)\left(S_{Tl}^2 - S_u^2\right) \cdot \left[T(\tilde{\alpha} - 1) - \left(T^6/48D_x\right)\left(S_{Tl}^2 - S_u^2\right)\right]$	Dickey–Fuller (1981, p. 1063)
(l) $\beta = 0$, $\alpha = 1$ in (3)	$Z(\Phi_3) = \left(S_u^2/S_{Tl}^2\right)\Phi_3 - \left(1/2S_{Tl}^2\right)\left(S_{Tl}^2 - S_u^2\right) \cdot \left[T(\tilde{\alpha} - 1) - \left(T^6/48D_x\right)\left(S_{Tl}^2 - S_u^2\right)\right]$	Dickey–Fuller (1981, p. 1063)

where

$$D_x = (T^2(T^2-1)/12)\Sigma y_{t-1}^2 - T(\Sigma t y_{t-1})^2 + T(T+1)\Sigma t y_{t-1}\Sigma y_{t-1} - (T(T+1)(2T+1)/6)(\Sigma y_{t-1})^2 = \det(X'X)$$

$$E_x = \left\{T^{-6}D_x + \tfrac{1}{12}\left(T^{-3/2}\Sigma y_{t-1}\right)^2\right\}^{1/2}$$

S_u^2 is a consistent estimator of $\sigma_u^2 = \lim_{T\to\infty} T^{-1}\Sigma_{t=1}^T \mathrm{E}(u_t^2)$, under the appropriate null hypothesis

S_{Tl}^2 is a consistent estimator, under the appropriate null hypothesis, of $\sigma^2 = \lim_{T\to\infty} T^{-1}\mathrm{E}(S_T^2)$, $S_T = \Sigma_{t=1}^T u_t$

2.4. Consistent estimation of σ^2 and σ_u^2

As stated in table 1, S_u^2 and S_{Tl}^2 refer to 'generic' estimators of σ_u^2 and σ^2, respectively, which are consistent under the null hypothesis of interest, i.e., data generating mechanisms (A) or (B). We now discuss the criteria for chosing particular estimators.

Under the data generating mechanism (A), the driftless case, the averaged sums of squared residuals under the null hypothesis, S_0^2, and under any of the three alternative regression models (1), (2) or (3), each of $\hat{S}^2$, S^{*2} and $\tilde{S}^2$ is a consistent estimator of σ_u^2, since the parameter estimates are themselves consistent. In the case of the data generating mechanism (B), the drift case, only S^{*2} and $\tilde{S}^2$ are consistent along with the estimator under the null hypothesis, $S_{0\mu}^2 = T^{-1}\sum_{t=1}^T (y_t - \bar{\mu} - y_{t-1})^2$ (where $\bar{\mu}$ is the OLS estimate of the constant term in a regression of $\Delta y_t = y_t - y_{t-1}$ on a constant only). In general, if the estimator under the alternative is chosen, the preferred one would be the one corresponding to the regression model used, because it corresponds to the variance estimate used in the construction of the t-statistics.

Matters are less simple for the consistent estimation of σ^2. The problem is equivalent to the consistent estimation of an asymptotic covariance matrix in the presence of weakly dependent and heterogeneously distributed innovations. This problem has recently been analysed by White and Domowitz (1984) [see also White (1984, ch. 6)]. Recall that $\sigma^2 = \lim_{T\to\infty} T^{-1}\mathrm{E}(S_T^2)$. Consider the finite sample variance of S_T,

$$\begin{aligned} V_T &= \mathrm{var}\left(T^{-1/2}S_T\right) \\ &= T^{-1}\sum_1^T \mathrm{E}\left(u_t^2\right) + 2T^{-1}\sum_{\tau=1}^{T-1}\sum_{t=\tau+1}^{T} \mathrm{E}(u_t u_{t-\tau}). \end{aligned}$$

We now denote l as the truncation lag parameter, and define the following approximant:

$$V_{Tl} = T^{-1}\sum_1^T \mathrm{E}\left(u_t^2\right) + 2T^{-1}\sum_{\tau=1}^{l}\sum_{t=\tau+1}^{T} \mathrm{E}(u_t u_{t-\tau}).$$

Intuition suggests that if we replace $\mathrm{E}(u_t^2)$ by its sample value, say u_t^2, we should be able to obtain a consistent estimate of σ^2. However, as in the Said and Dickey approach, we should take into consideration a number of lagged autocorrelations which increases with the sample size. Indeed, letting $u_t = \Delta y_t$, Phillips (1987a) proved that

$$S_{Tl}^2 = T^{-1}\sum_1^T u_t^2 + 2T^{-1}\sum_{\tau=1}^{l}\sum_{t=\tau+1}^{T} u_t u_{t-\tau} \tag{5}$$

is a consistent estimate of σ^2, under model (A), if the conditions of Assumption 1 are strengthened as follows:

Assumption 2

(a) The sequence $\{u_t\}_1^\infty$ satisfies the conditions of Assumption 1 and part (b) is replaced by the stronger moment condition $\sup_t \mathrm{E}|u_t|^{2\beta} < \infty$ for some $\beta > 2$.

(b) $l \uparrow \infty$ as $T \uparrow \infty$ such that $l = o(T^{1/4})$.

Just as there are choices for the consistent estimation of σ_u^2, there are also choices for the consistent estimation of σ^2. Indeed, under data generating mechanism (A), the driftless case, one can use the residuals under the null hypothesis, $u_{0t} = y_t - y_{t-1}(S_{Tl,0}^2)$ or under the various alternatives: $\hat{u}_t = y_t - \hat{\alpha} y_{t-1}(\hat{S}_{Tl}^2)$, $u_t^* = y_t - \mu^* - \alpha^* y_{t-1}(S^{*2}_{Tl})$ or $\tilde{u}_t = y_t - \tilde{\mu} - \tilde{\beta}(t - T/2) - \tilde{\alpha} y_{t-1}(\tilde{S}_{Tl}^2)$. Under the data generating mechanism (B), the drift case, only S_{Tl}^{*2}, $\tilde{S}_{Tl}^2$ and $S_{Tl,\mu}^2$ (which uses the residuals of a regression of Δy_t on a constant) are consistent. Again, the preferred estimator under the alternative hypothesis will depend on the regression model used.

The various choices concerning the estimators of σ^2, discussed above, are appropriate when considering the limiting distribution of the statistics under the null hypothesis of a unit root. However, when considering the behavior of the statistics under the alternative hypothesis of no unit root matters are less simple. Any choice described above will still lead to test statistic having the same asymptotic local power properties.[18] However, the consistency properties are different if we consider a fixed alternative.[19] Using the residuals estimated under the alternative hypothesis ($\hat{u}_t$, u_t^* or $\tilde{u}_t$ in the corresponding regression models) will lead to consistent tests under the assumptions stated above (Assumption 2). However, the test statistics will not be consistent against a fixed alternative if the residuals under the null hypothesis are used unless some very stringent, and in practice useless, conditions are imposed on the structure of the errors and the behavior of the truncation lag parameter l. For these reasons the recommendation is to use the corresponding regression residuals in the construction of the estimator of σ^2.

A problem with the estimator S_{Tl}^2, defined by (5) and any of the variations discussed above, is that it does not guarantee a nonnegative estimate in finite samples. Though this may appear to be a theoretical curiosity, it was found to be a problem in a substantial number of empirical series analyzed in section 3. One way to deal with this problem is to increase the truncation lag parameter, l, until a nonnegative estimate is obtained. This is not, however, a satisfactory solution.

[18] These properties are discussed in more details below.

[19] I am grateful to James Stock and Peter Phillips for helpful discussion on this issue.

Nevertheless, a nonnegative estimate can be constructed by weighting the sample lagged autocorrelations differently. Consider, for instance, the following estimator:

$$\sigma_{Tl}^2 = T^{-1}\sum_1^T u_t^2 + 2T^{-1}\sum_{\tau=1}^{l}\omega(\tau,l)\sum_{t=\tau+1}^{T} u_t u_{t-\tau},$$

where $\omega(\tau,l) = 1 - [\tau/(l+1)]$. This estimator was proposed by Newey and West (1987) in the context of the estimation of an asymptotic covariance matrix with dependent and heterogeneously distributed data. They also showed that σ_{Tl}^2 is a consistent estimator of σ^2 under the same conditions as S_{Tl}^2 (see Assumption 2) and that it was nonnegative by construction. In this context, σ_{Tl}^2 has a natural interpretation: 2π times the spectral density estimator of σ^2 at frequency zero, where a triangular lag window is used to smooth the estimated spectrum [see, for example, Priestley (1981, pp. 439–440)]. Of course, weights other than this one are possible and may be inspired by other choices of the lag window in the density estimate. Again, for reasons discussed above, the preferred estimator of σ^2 would use the estimated residuals under the alternative. We denote these estimators as $\hat{\sigma}_{Tl}^2$, σ_{Tl}^{*2} and $\tilde{\sigma}_{Tl}^2$ corresponding to regression models (1), (2) and (3), respectively.

Perron (1987a) discussed the finite sample properties of various estimators of the spectral density at the origin using a variety of windows (seventeen, five of which implied a nonnegative estimate) allowing for a range of truncation lag parameters. In that study analytical expressions are derived for the exact bias and mean-squared error. These are evaluated under a variety of data generating mechanisms. Though this analysis is, strictly speaking, not directly applicable to the estimators σ_{Tl}^2 when the regressions residuals are used (because these are estimated data and not realizations of a stochastic process), it can provide some guidelines as to the appropriate choice of the window and truncation lag parameter l.

The conclusions of this study can be summarized as follows: a) In general the properties of the estimators are more influenced by the choice of the truncation lag parameter l than by the choice of the window. Nevertheless some better and worse windows can be isolated. Among the more reliable windows which yield a positive definite estimate by construction are the Parzen window [$\omega(\tau,l) = 1 - 6(\tau/l)^2 + 6(\tau/l)^3$ if $0 < \tau/l < 1/2$ and $\omega(\tau,l) = 2(1-\tau/l)^3$ if $1/2 \le \tau/l \le 1$] and the 'Bohman' window [$\omega(\tau,l) = (1-\tau/l)\cos(\pi\tau/l) + \sin(\pi\tau/l)/\pi$]. b) The bias and mean squared error (MSE) are heavily influenced by the choice of the truncation lag parameter. When the variables are positively autocorrelated, a small value of l will minimize the MSE [e.g., MA(1) with a positive parameter]. When the variables are negatively autocorrelated, a large value of l will minimize the MSE [e.g., MA(1) with a negative parameter]. c) If the variables are nearly nonstationary, a large l is needed [e.g., AR(1) with a parameter near one]. d) If the variables have a near negative unit root, the estimators behave unreliably and in different ways [e.g., AR(1) with a parameter near minus one]. A recommendation from this study would be to do some preliminary investigation as to the nature of the correlation to determine the appropriate value of l. Another way to approach the problem is to check the sensitivity of the results to various values of l. This is done in the empirical applications of section 3.

2.5. Properties of the transformed statistics

The most interesting property of the new class of proposed test statistics concerns their asymptotic power under local alternatives. To this effect we consider a sequence of local alternatives of the form

$$\alpha = \exp(c/T) \sim 1 + c/T. \tag{6}$$

When $c = 0$, (6) reduces to the null hypothesis $\alpha = 1$; $c > 0$ corresponds to local explosive alternatives; and $c < 0$ corresponds to local stationary alternatives. The derivations of the noncentral asymptotic distribution of the original statistics and of Phillips' transformed statistics reveals the local asymptotic power of each class of tests.

The main result[20] is that the transformed statistics have the same noncentral limiting distribution, under the general case where the innovation sequence $\{u_t\}_1^\infty$ satisfies Assumption 2, as do the original statistics under the restricted case when the innovation sequence is i.i.d.. This result implies that the tests proposed here have the same asymptotic local power under general error structures as the original statistics have under i.i.d. errors. Nothing is lost by using the transformation when there is no need for them, i.e., when the errors are i.i.d..

Of course, the above property is asymptotic in nature and the finite sample properties need to be evaluated. Extensive Monte Carlo investigations have been presented by Perron (1986a, ch. 4), Phillips and Perron (1986) and Schwert (1987) for the case where the errors are a moving average of order one.

Schwert (1987) argued that tests for a unit root have seriously distorted actual sizes in finite samples when the true process was IMA(1, 1) with a moving average root close to minus unity. This problem is more severe for the Phillips–Perron tests than the Said–Dickey procedure. However, as shown by Phillips and Perron (1986), when the moving average parameter is positive, the Said–Dickey procedure yields a liberal test while the Phillips–Perron tests are both conservative and more powerful (and not much influenced by the choice of the truncation lag parameter).

[20] The proofs for statistics a and b (table 1) are in Phillips (1987c); for c, d, g and h see Phillips and Perron (1986); and for f, k and l see Perron (1986b).

To illustrate these issues we present, in table 2, the results of a small Monte Carlo experiment similar to the one in Phillips and Perron (1986). We used 1,000 replications of the model

$$y_t = \alpha y_{t-1} + e_t + \theta e_{t-1}, \tag{7}$$

with $\alpha = 1$ when calculating the exact size and $\alpha = 0.85$ when calculating the power. The sample size T is set at 200 and we used the Parzen window with the estimated residuals to construct the estimator σ_{Tl}^{*2}. The statistics considered are those associated with a regression that includes a fitted mean, $Z(\alpha^*)$, $Z(t_{\alpha^*})$ and Said–Dickey's t_{α^*}.

These results illustrate the potential gains available by using the Phillips–Perron statistics when θ is positive or zero; greater power and less influence of the particular choice of the truncation lag parameter. However, they clearly show the pitfalls of the Z-statistics when θ is negative (especially if its absolute value is large). Here the Said–Dickey procedure is more appropriate.

The usefulness of the Phillips–Perron approach would be undermined if economic variables were often characterized by the presence of a large negative moving average component. However, this does not appear to be the case as documented by the study of Nelson and Plosser (1982). They indeed concluded that most macroeconomic variables are well represented by IMA(1, 1) processes and they also show that the first-differences of the variables they considered have positive autocorrelations. It therefore appears that the most common case is that of positive moving average components. Nevertheless, Schwert (1987) presents some examples where the moving average component is negative, so care should be applied in some instances when interpreting the results.

This behavior of the test statistics under the null hypothesis, when θ is a large negative value, can be explained by noting that as θ approaches minus one the process (7) with $\alpha = 1$ becomes indistinguishable from a stationary process. The arguments used to develop the asymptotic distribution of the test statistics under the null hypothesis are valid for nonstationary series and can be expected to yield poor approximations to 'nearly stationary series'. In analogy to the new asymptotic distribution of 'near-integrated processes' [see Phillips (1986) and Perron (1987c)], an interesting topic for future research would be to enquire about an asymptotic distribution for nonstationary series that are 'nearly stationary'.

2.6. *A testing strategy*

A testing strategy encompasses the choice of the appropriate regression model, the statistics to be used and, when possible, the choice of the sample of data to be analyzed.

Table 2

Monte Carlo simulations on the size and power of the $Z(\alpha^*)$, $Z(t_{\alpha^*})$ and t_{α^*} tests; $T = 200$; $l = 2, 6, 10, 14, 20$.

	Size					Power ($\alpha = 0.85$)				
	2	6	10	14	20	2	6	10	14	20
					Said–Dickey t_{α^*}					
$\theta = 0.0$	0.059	0.066	0.072	0.077	—	0.975	0.857	0.730	0.634	—
$\theta = 0.5$	0.034	0.050	0.064	0.067	—	0.908	0.850	0.714	0.604	—
$\theta = 0.8$	0.025	0.044	0.056	0.067	—	0.807	0.767	0.655	0.561	—
$\theta = -0.2$	0.055	0.062	0.067	0.069	—	0.982	0.872	0.754	0.631	—
$\theta = -0.5$	0.130	0.074	0.072	0.085	—	1.00	0.936	0.790	0.671	—
$\theta = -0.8$	0.738	0.180	0.086	0.076	—	1.00	1.00	0.981	0.877	—
					$Z(\alpha^*)$					
$\theta = 0.0$	0.058	0.063	0.066	0.064	0.064	1.00	0.999	0.999	0.999	1.00
$\theta = 0.5$	0.004	0.022	0.022	0.018	0.013	0.932	0.988	0.981	0.971	0.946
$\theta = 0.8$	0.005	0.026	0.027	0.025	0.015	0.894	0.989	0.977	0.955	0.895
$\theta = -0.2$	0.139	0.094	0.106	0.115	0.135	1.00	1.00	1.00	1.00	1.00
$\theta = -0.5$	0.585	0.363	0.426	0.488	0.551	1.00	1.00	1.00	1.00	1.00
$\theta = -0.8$	0.998	0.977	0.987	0.993	0.996	1.00	1.00	1.00	1.00	1.00
					$Z(t_{\alpha^*})$					
$\theta = 0.0$	0.068	0.066	0.070	0.065	0.066	0.997	0.998	0.998	0.999	0.999
$\theta = 0.5$	0.018	0.028	0.030	0.026	0.021	0.819	0.952	0.933	0.898	0.806
$\theta = 0.8$	0.023	0.035	0.035	0.031	0.029	0.758	0.944	0.917	0.861	0.725
$\theta = -0.2$	0.121	0.096	0.107	0.119	0.129	1.00	1.00	1.00	1.00	1.00
$\theta = -0.5$	0.546	0.376	0.415	0.469	0.527	1.00	1.00	1.00	1.00	1.00
$\theta = -0.8$	0.996	0.976	0.984	0.990	0.994	1.00	1.00	1.00	1.00	1.00

In a recent study, Dickey, Bell and Miller (1986) advocated the use of regression model (2) at the outset. First, they argue that it avoids the low power of the statistics from regression model (3) [under model (A)] and the low power of the statistics from regression model (1) caused by a nonzero mean. Secondly, they recognize that the statistics from regression model (2) may have low power when a deterministic linear trend is present, but argue that 'since differencing removes a linear trend, the lower power of [these statistics], in this case, may actually be more comforting than alarming' (p. 18). While it is true that differencing removes a linear trend, which for forecasting purposes may be adequate, the use of the statistics from regression model (2) precludes any test of the unit root hypothesis against the hypotheses that the series is stationary around a linear trend. This conclusion is formally demonstrated in the following theorem proved in appendix 1.

Theorem 1. If $\{y_t\}_0^\infty$ *is a stationary series around a deterministic linear trend, then as* $T \uparrow \infty$:

$$(a)\quad T(\alpha^* - 1) \underset{p}{\rightarrow} 0, \qquad (b)\quad t_{\alpha^*} \underset{p}{\rightarrow} 0.$$

Theorem 1 shows that both the normalized estimator of α^* and its t-statistic tend in the limit to zero. The consequence is that the use of the statistics from regression model (2) cannot distinguish a stationary process around a linear trend from a process with a unit root. Indeed, rejection of the null hypothesis of a unit root is unlikely if the series is stationary around a linear trend and becomes impossible as the sample size increases. Therefore, regression model (2) is inappropriate if a plausible alternative is that the series is trend stationary. The intuition for this result is straightforward. Suppose a series shows a definite tendency to have an increase in its mean over time and a regression is used with only a constant and a lagged dependent variable. The only way to capture this increase is to make the constant become a drift term which occurs when the autoregressive parameter is set at one. The theorem not only formalizes this intuition but shows that the convergence of the autoregressive parameter to one is fast enough so that the usual test statistics will not reject the unit root, even asymptotically.

This conclusion, along with the inadequacy of regression model (2) in the presence of a drift, highlights the significance of regression model (3). Hypothesis testing should begin with statistics derived from model (3).[21]

The strategy proposed here is as follows. Estimate regression model (3) and use the statistics $Z(\tilde{\alpha})$, $Z(t_{\tilde{\alpha}})$ and $Z(\Phi_3)$ to assess whether there is evidence

[21]For example, Kleidon (1986, p. 993) finds that the unit root hypothesis is rejected for real earnings using regression model (3) but not rejected using regression model (2). This is well explained by theorem 1 which makes clear that his preferences for concluding that real earnings are nonstationary with a unit root is inappropriate. This conclusion is also obtained in section 3.

for rejecting the null hypothesis of a unit root.[22] If rejection is possible, there is no need to go further. If one cannot reject, this may be due to the poor power properties of these statistics compared to those of regression model (2). However, before implementing statistics based on this model, it must be verified that the drift μ is zero, since they are not invariant with respect to μ. So, we use the statistic $Z(\Phi_2)$ [$Z(t_{\tilde{\mu}})$ is also possible, but it is to be noted that it is not invariant with respect to the initial observation y_0 in finite samples]; if $Z(\Phi_2)$ suggests that the null hypothesis $(\mu, \beta, \alpha) = (0, 0, 1)$ cannot be rejected, then the use of the statistics from regression model (2) may be more appropriate. The test is then carried through via the statistics $Z(\alpha^*)$, $Z(t_{\alpha^*})$ and $Z(\Phi_1)$. If the series has zero mean, then regression model (1) may be more appropriate.

As far as the choice of the sample of data is concerned, it should be noted that more observations do not necessarily lead to tests having higher power if there is a change in the sampling interval. This feature has been documentated both by theoretical considerations and Monte Carlo experiments in Perron (1987b) and Shiller and Perron (1985), where it was shown that the power of a test of the random walk hypothesis depends much more on the span of the data than on the number of observations per se. Theoretically, Perron (1987b) showed that tests of the random walk hypothesis are consistent against a fixed disjoint alternative in terms of the underlying continuous time process (whether explosive or stationary) if and only if the span of the data increases with the sample size. Of course, for a given span, more observations lead to higher power but the power tends to level off at a value which is an increasing function of the span. Similarly, a longer span for a given number of observations leads to a higher power.

With macroeconomic data, the relevant alternative being mean reversion over a period of an order similar to business cycles, a long span of annual data is to be preferred to a shorter span with, say, quarterly or monthly data even if the latter affords a greater number of observations.

Nevertheless, one must bear in mind that any existing test of the unit root hypothesis has quite a low power against relevant stationary alternatives with a root close to unity (with the sample of data usually available in economics). This is a problem that cannot be avoided. A root close to unity would imply that mean reversion occurs over very long periods of time. Again, this highlights the importance of the span of the available data rather than the number of observations per se.

[22] $Z(\Phi_3)$ should also be used because the limiting distributions of $Z(\tilde{\alpha})$ and $Z(t_{\tilde{\alpha}})$ are not invariant to the trend parameter β, under the null hypothesis of a unit root. $Z(t_{\tilde{\beta}})$ is not invariant with the respect to μ, the drift, so can only be used if $Z(\Phi_2)$ suggests the absence of a drift.

3. Some empirical evidence

Nelson and Plosser (1982) stressed the importance of determining whether a time series was stochastic nonstationary. As argued in the introduction, the conclusions have deep implications for economic theory. Using the Dickey–Fuller (1979) approach, they found strong support for the hypothesis that, in fact, most macroeconomic time series are characterized by the presence of a unit root; the nonstationarity is stochastic as opposed to deterministic (in the form of a linear trend, say). In this section, we reassess their findings using the methodology described in the previous section. We also present some empirical evidence on a variety of other macroeconomic series that have been the subject of some debate concerning their stationarity.

All the results reported in this section are based on the Newey–West estimator of σ^2, σ^2_{Tl}, and use the residuals under the alternative corresponding to the relevant regression model.

3.1. The Nelson–Plosser results revisited

Nelson and Plosser analyzed a set of fourteen macroeconomic time series and found that all of them, except the unemployment rate series, were characterized by stochastic nonstationarity. We use exactly the same data set and, as they do, we analyze the logarithm of these series (except for the interest rate series) instead of the level to account for the fact that there is a tendency for the dispersion of the series to increase with the absolute level.[23]

Table 3 presents the results obtained using regression model (3). The null hypothesis is that $\alpha = 1$ and $\beta = 0$. This can be tested using three different statistics, $Z(\tilde{\alpha})$ $(\alpha = 1)$, $Z(t_{\tilde{\alpha}})$ $(\alpha = 1)$, and $Z(\Phi_3)$ $(\alpha = 1$ and $\beta = 0)$. Overall, the results strongly support the conclusions reached by Nelson and Plosser. The three statistics mentioned above are not significant at the 5% level (for any value of the truncation lag parameter l) for all the series examined except the unemployment rate. For the unemployment rate series $Z(\tilde{\alpha})$, $Z(t_{\tilde{\alpha}})$ and $Z(\Phi_3)$ are significant at the 5% level for $l = 1$ and $l = 4$ and at the 10% level for higher values of l. For the industrial production series, $Z(\tilde{\alpha})$ and $Z(t_{\tilde{\alpha}})$ are significant at the 10% level for $l = 1, 4$. However, $Z(\Phi_3)$ is not significant at the 10% level for any value of l. The evidence is, therefore, rather marginal for the industrial production series, marginally in favor of not rejecting the null hypothesis of a unit root.

The statistics $Z(\Phi_2)$ (for the null hypothesis that $\mu = 0$, $\beta = 0$ and $\alpha = 1$) is not significant at the 5% level for the following series: real per capita GNP, the

[23]'The tendency of economic time series to exhibit variation that increases in mean and dispersion in proportion to absolute level, motivates the transformation to natural logs and the assumption that trends are linear in the transformed data' [Nelson and Plosser 1982, p. 141)].

Table 3

Tests for autoregressive unit roots in Nelson–Plosser series.[a]

	$l=1$	$l=4$	$l=7$	$l=10$	$l=1$	$l=4$	$l=7$	$l=10$	$l=1$	$l=4$	$l=7$	$l=10$
	Real GNP ($\tilde{\alpha}=0.876$)				*Nominal GNP* ($\tilde{\alpha}=0.935$)				*Real per capita GNP* ($\tilde{\alpha}=0.868$)			
$Z(\tilde{\alpha})$	−9.96	−10.52	−8.85	−7.15	−5.77	−6.69	−6.67	−6.82	−10.46	−11.04	−9.45	−7.77
$Z(t_{\tilde{\alpha}})$	−2.35	−2.40	−2.23	−2.03	−1.68	−1.82	−1.81	−1.83	−2.43	−2.49	−2.33	−2.15
$Z(\Phi_3)$	2.90	3.03	2.65	2.28	1.51	1.73	1.72	1.76	3.10	3.09	2.86	2.50
$Z(\Phi_2)$	5.25	5.14	5.55	6.34	5.46	5.00	5.01	4.95	3.06	3.23	3.03	3.07
	Industrial production ($\tilde{\alpha}=0.841$)				*Employment* ($\tilde{\alpha}=0.889$)				*Unemployment rate* ($\tilde{\alpha}=0.748$)			
$Z(\tilde{\alpha})$	−19.08	−19.06	−16.79	−15.77	−11.62	−12.36	−10.88	−9.68	−23.33	−22.25	−21.02	−19.42
$Z(t_{\tilde{\alpha}})$	−3.24	−3.24	−3.06	−2.98	−2.51	−2.58	−2.43	−2.31	−3.64	−3.57	−3.48	−3.37
$Z(\Phi_3)$	5.18	5.17	4.60	4.34	3.13	3.32	2.95	2.65	6.46	6.18	5.87	5.46
$Z(\Phi_2)$	10.01	10.02	10.63	11.01	5.71	5.60	5.86	6.19	4.31	4.12	3.91	3.64
	GNP deflator ($\tilde{\alpha}=0.933$)				*Consumer prices* ($\tilde{\alpha}=0.987$)				*Wages* ($\tilde{\alpha}=0.938$)			
$Z(\tilde{\alpha})$	−7.29	−9.60	−10.29	−10.70	−2.91	−4.16	−4.43	−4.50	−6.38	−8.01	−8.37	−8.60
$Z(t_{\tilde{\alpha}})$	−2.09	−2.34	−2.41	−2.45	−1.04	−1.29	−1.34	−1.36	−1.80	−2.02	−2.06	−2.09
$Z(\Phi_3)$	2.87	3.24	3.37	3.45	1.30	1.41	1.45	1.46	1.75	2.13	2.21	2.27
$Z(\Phi_2)$	4.91	4.34	4.26	4.23	2.12	1.89	1.87	1.87	6.75	5.85	5.72	5.64
	Real wages ($\tilde{\alpha}=0.871$)				*Money stock* ($\tilde{\alpha}=0.949$)				*Velocity* ($\tilde{\alpha}=0.941$)			
$Z(\tilde{\alpha})$	−10.51	−10.73	−8.86	−7.72	−6.67	−10.18	−10.89	−10.46	−6.68	−5.71	−4.63	4.28
$Z(t_{\tilde{\alpha}})$	−2.53	−2.55	−2.37	−2.26	−1.85	−2.27	−2.35	−2.30	−1.79	−1.65	−1.48	−1.42
$Z(\Phi_3)$	3.31	3.36	2.95	2.72	1.71	2.57	2.76	2.66	2.95	2.93	3.01	3.07
$Z(\Phi_2)$	7.21	7.12	8.16	9.20	15.81	11.27	10.76	11.06	2.88	3.01	3.29	3.43
	Interest rate ($\tilde{\alpha}=1.075$)				*Common stock prices* ($\tilde{\alpha}=0.921$)							
$Z(\tilde{\alpha})$	4.98	3.32	2.78	2.41	−9.67	−8.97	−7.26	−7.56				
$Z(t_{\tilde{\alpha}})$	1.74	0.98	0.78	0.66	−2.20	−2.12	−1.90	−1.94				
$Z(\Phi_3)$	3.99	2.24	1.91	1.74	2.92	2.78	2.48	2.53				
$Z(\Phi_2)$	3.66	2.21	2.93	1.77	2.84	2.82	2.85	2.83				

[a]Critical values	10%	5%	2.5%	1%
$Z(\tilde{\alpha})$	−18.3	−21.8	−25.1	−29.5
$Z(t_{\tilde{\alpha}})$	−3.12	−3.41	−3.66	−3.96
$Z(\Phi_3)$	5.34	6.25	7.16	8.27
$Z(\Phi_2)$	4.03	4.68	5.31	6.09

Sources: Fuller (1976) and Dickey–Fuller (1981).
The estimators of σ_u^2 and σ^2 are $\tilde{S}^2$ and $\tilde{\sigma}_{Tl}^2$, respectively.
All variables are in logarithmic form except for the interest rate.

Table 4

Tests for a unit root using regression model (2).[a]

	$l=1$	$l=4$	$l=7$	$l=10$	$l=1$	$l=4$	$l=7$	$l=10$	$l=1$	$l=4$	$l=7$	$l=10$
	Real per capita GNP ($\alpha^* = 0.998$)				*Unemployment rate* ($\alpha^* = 0.754$)				*Consumer prices* ($\alpha^* = 1.010$)			
$Z(\alpha^*)$	−0.50	−0.53	−0.26	−0.03	−22.79	−21.78	−20.51	−18.86	0.59	0.10	−0.04	−0.10
$Z(t_{\alpha^*})$	−0.29	−0.31	−0.17	−0.02	−3.58	−3.51	−3.42	−3.30	0.34	0.05	−0.02	−0.05
$Z(\Phi_1)$	1.42	1.39	1.71	2.11	6.04	5.91	5.69	5.27	1.82	1.14	0.99	0.93
	Velocity rate ($\alpha^* = 0.96$)				*Interest rate* ($\alpha^* = 1.076$)				*Common stock prices* ($\alpha^* = 1.003$)			
$Z(\alpha^*)$	−4.02	−3.74	−3.46	−3.34	4.81	2.57	1.55	0.64	−0.02	0.26	0.74	0.77
$Z(t_{\alpha^*})$	−2.33	−2.40	−2.54	−2.63	1.58	0.69	0.39	0.15	−0.01	0.14	0.47	0.50
$Z(\Phi_1)$	4.05	4.52	5.28	5.74	2.52	0.57	−0.06	−0.55	1.28	1.54	2.18	2.24

[a]Critical values	10%	5%	2.5%	1%
$Z(\alpha^*)$	−11.3	−14.1	−16.9	−20.7
$Z(t_{\alpha^*})$	− 2.57	−2.86	−3.12	−3.43
$Z(\Phi_1)$	3.78	4.59	5.38	6.43

Sources: Fuller (1976) and Dickey and Fuller (1981).
The estimators of σ_u^2 and σ^2 are S^{*2} and σ_{Tl}^{*2}, respectively.
All variables are in logarithmic form except for the interest rate.

unemployment rate, consumer prices, velocity, interest rate and common stock prices. This suggests the absence of a drift for these series ($\mu = 0$) and that a more powerful set of test statistics can be obtained from regression model (2). Results for the statistics $Z(\alpha^*)$, $Z(t_{\alpha^*})$ and $Z(\Phi_1)$ are presented in table 4 for these series. The conclusion of the presence of a unit root is confirmed for real per capita GNP, consumer prices, interest rate and common stock prices. For the velocity series, $Z(\alpha^*)$ is not significant at the 10% level for any l, while $Z(t_{\alpha^*})$ is significant at the 10% level only for $l = 10$ and $Z(\Phi_1)$ is significant at the 5% level with $l = 7, 10$. While the evidence is mixed for this series, it does not tend to support the idea that the velocity series is stationary. For the unemployment rate series, $Z(\alpha^*)$ and $Z(t_{\alpha^*})$ are significant at the 1% level which shows how the statistics of regression model (2) can be more powerful than those of regression model (3) in the absence of an increase in the level of the series.

The conclusions with regard to the Nelson–Plosser series are as follows: the unemployment rate series and possibly the industrial production index are stationary around a linear trend (albeit a zero trend for the former). The following series are characterized by the presence of a unit root without a drift: real per capita GNP, consumer prices, velocity, interest rate and common stock prices. The remaining series have a unit root with a drift.

3.2. *Further evidence on additional series*

Table 5 presents the estimated statistics from regression model (3) for a number of variables for which the unit root hypothesis has been advanced. The exact definition and sources of each series is given in appendix 2.

Consider first the case of aggregate consumption which, as argued in Hall (1978), should behave as a martingale. Series 1 through 4 consider respectively aggregate values for nominal consumption, real consumption and their per capita counterparts. For all these series, there is no evidence to reject the hypothesis that consumption is a random walk with drift.

The case for dividends and earnings has recently been advanced by Kleidon (1986), among others. Series 5 and 6 show that nominal aggregate dividends and earnings are nonstationary with a unit root. However, real dividend (series 7) and real earnings (series 8) are stationary around a significant linear trend when the deflator used is the producer price index. To check the robustness of these results, we examined the same nominal series deflated by the price deflator for personal expenditures on nondurables and services (series 9 and 10). We still reject the hypothesis of a unit root at the 5% level for earnings but only at the 10% for dividends. This difference may result from the fact that only 91 observations were available for the consumption deflator as compared to 114 for the producer price index, thereby suggesting a loss in power. As

Table 5

Tests for autoregressive unit roots for various aggregate series.[a]

	$l=1$	$l=4$	$l=7$	$l=10$	$l=1$	$l=4$	$l=7$	$l=10$	$l=1$	$l=4$	$l=7$	$l=10$
	1) *Nominal consumption* ($\tilde{\alpha}=0.982$) (1889–1979)				2) *Real consumption* ($\tilde{\alpha}=0.907$) (1889–1979)				3) *Nominal consumption per capita* ($\tilde{\alpha}=0.982$) (1889–1979)			
$Z(\tilde{\alpha})$	−3.03	−5.15	−5.96	−6.51	−7.93	−9.47	−9.73	−9.43	−3.02	−5.13	−5.81	−6.26
$Z(t_{\tilde{\alpha}})$	−0.94	−1.34	−1.47	−1.56	−2.16	−2.33	−2.35	−2.32	−0.92	−1.32	−1.43	−1.50
$Z(\Phi_3)$	1.93	1.96	2.04	2.11	2.29	2.68	2.74	2.67	2.51	2.38	2.41	2.44
$Z(\Phi_2)$	18.38	13.49	12.34	11.68	27.62	23.01	22.41	23.11	11.17	8.43	7.88	7.58
	4) *Real consumption per capita* ($\tilde{\alpha}=0.885$) (1889–1979)				5) *Dividend* ($\tilde{\alpha}=0.930$) (1871–1984)				6) *Earnings* ($\tilde{\alpha}=0.847$) (1871–1984)			
$Z(\tilde{\alpha})$	−9.54	−11.78	−12.20	−11.79	−9.90	−8.74	−6.41	−5.90	−18.43	−16.48	−14.25	−13.90
$Z(t_{\tilde{\alpha}})$	−2.13	−2.37	−2.42	−2.38	−2.25	−2.11	−1.82	−1.74	−3.16	−3.00	−2.81	−2.78
$Z(\Phi_3)$	2.24	2.80	2.91	2.81	3.51	3.35	3.19	3.20	5.32	4.88	4.40	4.32
$Z(\Phi_2)$	9.75	8.62	8.47	8.62	3.96	4.07	4.65	4.88	4.30	4.10	3.92	3.90
	7) *Real dividend* (*PPI*) ($\tilde{\alpha}=0.796$) (1871–1984)				8) *Real earnings* (*PPI*) ($\tilde{\alpha}=0.694$) (1871–1984)				9) *Real dividend* (*PC*) ($\tilde{\alpha}=0.82$) (1889–1979)			
$Z(\tilde{\alpha})$	−26.86	−25.70	−24.57	−22.42	−37.50	−35.59	−32.30	−30.87	−19.46	−19.23	−17.60	−15.06
$Z(t_{\tilde{\alpha}})$	−3.85	−3.78	−3.70	−3.55	−4.67	−4.57	−4.39	−4.32	−3.26	−3.24	−3.12	−2.91
$Z(\Phi_3)$	7.30	7.02	6.72	6.18	10.67	10.19	9.37	9.01	5.21	5.15	4.74	4.09
$Z(\Phi_2)$	5.20	5.03	4.85	4.53	7.33	7.02	6.50	6.28	3.76	3.73	3.48	3.11
	10) *Real earnings* (*PC*) ($\tilde{\alpha}=0.720$) (1889–1979)				11) *Dividend/price ratio* ($\tilde{\alpha}=0.689$) (1871–1983)				12) *Earning/price ratio* ($\tilde{\alpha}=0.699$) (1871–1983)			
$Z(\tilde{\alpha})$	−27.74	−26.67	−23.03	−21.48	−33.40	−34.19	−35.48	−39.75	33.04	−33.99	−33.25	−35.58
$Z(t_{\tilde{\alpha}})$	−3.97	−3.90	−3.66	−3.56	−4.47	−4.51	−4.58	−4.80	−4.42	−4.47	−4.43	−4.56
$Z(\Phi_3)$	7.65	7.38	6.45	6.05	9.68	9.88	10.20	11.27	9.50	9.74	9.55	10.14
$Z(\Phi_2)$	5.30	5.13	4.55	4.30	6.46	6.59	6.81	7.52	6.34	6.49	6.37	6.76

	13) *Price (JUL) + dividend* ($\tilde{\alpha} = 0.876$) (1871–1983)				14) *Price index + dividend* ($\tilde{\alpha} = 0.907$) (1889–1979)				15) *Price (JAN) + dividend* ($\tilde{\alpha} = 0.912$) (1871–1984)			
$Z(\tilde{\alpha})$	−13.00	−12.08	−10.62	−11.73	−9.94	−9.42	−8.15	−8.56	−10.85	−9.25	−7.91	−8.17
$Z(t_{\tilde{\alpha}})$	−2.63	−2.55	−2.40	−2.51	−2.34	−2.28	−2.14	−2.19	−2.35	−2.18	−2.02	−2.05
$Z(\Phi_3)$	3.94	3.75	3.46	3.68	2.88	2.76	2.47	2.56	3.27	2.96	2.73	2.77
$Z(\Phi_2)$	3.65	3.60	3.58	3.59	3.01	3.00	3.01	3.00	3.38	3.39	3.48	3.45
	16) *Price (JUL) + dividend (real)* ($\tilde{\alpha} = 0.813$) (1871–1983)				17) *Real SP500 JUL (PPI)* ($\tilde{\alpha} = 0.810$) (1871–1983)				18) *Real SP500 JUL (PC)* ($\tilde{\alpha} = 0.811$) (1889–1979)			
$Z(\tilde{\alpha})$	−18.82	−20.63	−21.98	−22.83	−19.10	−20.94	−22.31	−23.28	−15.59	−17.00	−17.50	−18.58
$Z(t_{\tilde{\alpha}})$	−3.24	−3.38	−3.47	−3.53	−3.27	−3.41	−3.50	−3.57	−2.93	−3.04	−3.08	−3.17
$Z(\Phi_3)$	5.10	5.57	5.91	6.13	5.19	5.66	6.01	6.25	4.12	4.48	4.61	4.89
$Z(\Phi_2)$	3.61	3.89	4.11	4.25	3.65	3.95	4.17	4.32	2.80	3.04	3.13	3.31
	19) *Real SP500 JAN (PPI)* ($\tilde{\alpha} = 0.881$) (1871–1984)				20) *Real SP500 JAN (PC)* ($\tilde{\alpha} = 0.876$) (1889–1979)				21) *Price index + dividend (real, PC)* ($\tilde{\alpha} = 0.892$) (1889–1979)			
$Z(\tilde{\alpha})$	−14.86	−15.15	−15.96	−16.21	−12.37	−12.57	−12.79	−13.12	−11.36	−11.91	−11.81	−11.96
$Z(t_{\tilde{\alpha}})$	−2.82	−2.84	−2.91	−2.93	−2.54	−2.56	−2.58	−2.61	−2.42	−2.47	−2.46	−2.48
$Z(\Phi_3)$	3.93	4.00	4.21	4.27	3.14	3.19	3.25	3.33	2.86	3.00	2.98	3.01
$Z(\Phi_2)$	2.87	2.91	3.03	3.07	2.16	2.19	2.23	2.28	1.99	2.08	2.07	2.09
	22) *Price index + dividend (real, PPI)* ($\tilde{\alpha} = 0.889$) (1889–1979)				23) *Population* ($\tilde{\alpha} = 0.971$) (1889–1979)							
$Z(\tilde{\alpha})$	−11.16	−11.75	−12.39	−12.40	−3.12	−4.34	−5.25	−5.83				
$Z(t_{\tilde{\alpha}})$	−2.40	−2.46	−2.53	−2.53	−2.04	−1.99	−2.05	−2.09				
$Z(\Phi_3)$	2.81	2.96	3.13	3.13	11.66	6.67	5.49	5.08				
$Z(\Phi_2)$	1.96	2.05	2.16	2.16	213.16	105.19	77.59	66.40				

[a]See table 3 for the critical values.
The estimators of σ_u^2 and σ^2 are $\tilde{S}^2$ and $\tilde{\sigma}_{Tl}^2$, respectively.
All variables are in logarithmic form.

shown by series 11 and 12, the dividend–price ratio and the earnings–price ratio are stationary.

These results are broadly consistent with the existing literature. Both Kleidon (1986) and Campbell and Shiller (1986) rejected the unit root with a drift hypothesis against a trend-stationary alternative for the logarithm of real earnings (using the producer price index as the deflator). Kleidon argued that the pre-1926 data were unreliable and when using the post-1926 sample only he could not reject the null hypothesis of a unit root. Campbell and Shiller (1987), in contrast, found that the level of real dividend and real earnings were better represented by a unit root with drift. The results on the logarithm of the dividend– and earnings–price ratios agree with those of Campbell and Shiller (1986) and confirm an interesting puzzle noted by these authors. Indeed, if the logarithm of stock prices is integrated of order one and the logarithm of dividend is trend-stationary, then the logarithm of the dividend–price ratio should also be integrated to order one. More work is needed to interpret these results.

To analyze the behavior of stock market prices, three series were available: The Standard and Poors 500 annual index (1889–1979) (series 14, 21, 22), the S&P 500 value for January (1871–1984) (series 15, 19, 20) and the S&P 500 value for July (1871–1983) (series 13, 16, 17, 18). Consider first the nominal series obtained by adding price and dividend (series 13, 14, 15). With any stock price variable there is no evidence to reject the hypothesis of a unit root without drift. When we examine the latter variable deflated by a price index, the results are mixed. First, when the S&P 500 July series is deflated by the producer price index (series 16, 17), there is evidence to reject the hypothesis of a unit root at the 5% level. [This result is mainly due to the fact that the real price series (without dividend) is stationary (see series 17).] However, the conclusion is not robust to using a consumption price deflator (series 18). The discrepency may be due to the fact that the producer price index uses solely the January value of each year whereas the consumption deflator is an annual average index.

To investigate the robustness of the above result, we examined real price and dividend series using the S&P 500 January series. When the latter series is deflated by the producer price index (series 19), there remains no evidence to reject the hypothesis of a unit root. The same result holds when deflating by the consumption deflator (see series 20). The same conclusion (of a nonstationary series with a unit root) holds using the annual S&P 500 index whether deflated by the producer price index or the consumption deflator (series 21, 22). It, therefore, appears that misalignment of the data can cause a series to appear stationary around a linear trend.

Finally, note that the population series itself exhibits stochastic nonstationarity (series 23). Since the $Z(\Phi_3)$ statistic is below its 5% critical value for many series, more powerful tests could be obtained using regression model

(2). In such cases, these statistics were computed but are not reported here, since the conclusions are not affected.

From the evidence presented in this section, we can conclude, that the consumption (whether real or per capita) and population series are nonstationary with a unit root and drift. The real stock price series also contains a unit root but no drift. Finally, the real dividend and earning series are stationary around a linear deterministic trend.

4. Conclusions

This paper has presented a new approach to hypothesis testing in time series regression with a unit root. The approach has several interesting advantages over currently existing procedures: 1) it is valid for very general time series models including most finite-order ARMA and ARMAX systems; 2) it incorporates the effects of correlation of the first differences on the statistics in a nonparametric way, and thereby reduces the need to estimate nuisance parameters; 3) the test statistics are relatively easy to construct, requiring only OLS regression estimates, the calculations of moments of the data and an estimator of the variances; 4) tests of hypotheses require critical values which have already been tabulated by Dickey and Fuller; 5) finally, the tests have the interesting property that their asymptotic local power, under general conditions on the innovation structure, is the same as the one obtained under i.i.d. conditions by the Dickey–Fuller approach. For these reasons, we think that the test statistics proposed have a considerable practical interest.

We have also examined a number of macroeconomic time series and tested the presence of stochastic nonstationarity. Our findings confirm the conclusions reached by Nelson and Plosser. It appears, indeed, that the nonstationarity of most macroeconomic variables is best construed as stochastic rather than deterministic. There are interesting exceptions, however, such as earnings, dividends and the unemployment rate.

Appendix 1

Theorem 1 is a straightforward corrolary to the following lemma (in this appendix $\sum$ denotes $\sum_1^T$):

Lemma A.1. If $\{y_t\}_0^T$ is a stationary series around a linear deterministic trend, then as $T \uparrow \infty$:

(a) $T^2(\alpha^* - 1) \xrightarrow{D} 12(\beta X - \sigma_u^2 + \rho_1)/\beta^2,$

(b) $T^{1/2} t_{\alpha^*} \xrightarrow{D} \sqrt{12}\,(\beta X - \sigma_u^2 + \rho_1)/\beta(\beta^2 + 2(\sigma_u^2 - \rho_1))^{1/2},$

where $\overset{D}{\to}$ *denotes 'asymptotically distributed as'*, $\sigma_u^2 = \lim_{T\to\infty} T^{-1}\sum \mathrm{E}(u_t^2)$, $X = \lim_{T\to\infty} \frac{1}{2}(u_T + u_0)$, and $\rho_1 = \lim_{T\to\infty} T^{-1}\sum \mathrm{E}(u_t u_{t-1})$.

Proof. If $\{y_t\}_0^\infty$ is stationary around a linear deterministic trend, then we can write it as

$$y_t = \mu + \beta t + u_t, \tag{A.1}$$

where u_t is a stationary process which satisfies the conditions of Assumption 1. For example, we may have $a(L)u_t = b(L)e_t$ with $e_t \sim$ i.i.d. $\mathrm{N}(0, \sigma^2)$ and $a(L)$ and $b(L)$ finite-order polynomials. Without loss of generality, we may set $\mu = 0$. Now

$$\alpha^* = \sum (y_t - \bar{Y}) y_{t-1} \left\{ \sum (y_{t-1} - \bar{Y}_{-1})^2 \right\}^{-1},$$

and

$$T^2(\alpha^* - 1) = \left\{ T^{-3} \sum (y_{t-1} - \bar{Y}_{-1})^2 \right\}^{-1} \times \left\{ T^{-1} \sum y_{t-1}(y_t - y_{t-1}) - T^{-2}(y_T - y_0) \sum y_{t-1} \right\}.$$

Consider the denominator; from (A.1):

$$T^{-3} \sum (y_{t-1} - \bar{Y}_{-1})^2 = T^{-3} \sum y_{t-1}^2 - \left(T^{-2} \sum y_{t-1} \right)^2,$$

(i)
$$\begin{aligned} T^{-3} \sum y_{t-1}^2 &= T^{-3} \sum \left(\beta(t-1) + u_{t-1} \right)^2 \\ &= T^{-3} \sum \left(\beta^2 t^2 - 2\beta^2 t + \beta^2 + 2\beta(t-1)u_{t-1} + u_{t-1}^2 \right) \\ &= T^{-3}\beta^2 \sum t^2 + \mathrm{o}(1) \\ &\to \beta^2/3, \end{aligned}$$

since $\sum u_{t-1} = \mathrm{O}(T^{1/2})$, $\sum t u_t = \mathrm{O}(T^{3/2})$ and $\sum u_{t-1}^2 = \mathrm{O}(T)$ [see Phillips and Perron (1986)].

Similarly,

(ii)
$$\begin{aligned} T^{-2} \sum y_{t-1} &= T^{-2} \sum \left(\beta(t-1) + u_{t-1} \right) \\ &= T^{-2}\beta \sum t + \mathrm{o}(1) \\ &\to \beta/2. \end{aligned}$$

Therefore,

$$T^{-3}\sum\left(y_{t-1}-\bar{Y}_{-1}\right)^2 \to \beta^2/12.$$

Now consider the numerator:

(iii) $T^{-1}\sum y_{t-1}(y_t - y_{t-1})$

$$= T^{-1}\sum(\beta t - \beta + u_{t-1})(\beta + u_t - u_{t-1})$$

$$= T^{-1}\sum\left(\beta^2 t + \beta t u_t - \beta t u_{t-1} - \beta^2 - \beta u_t + \beta u_{t-1}\right.$$

$$\left. + \beta u_{t-1} + u_{t-1}u_t - u_{t-1}^2\right),$$

(iv) $T^{-2}(y_T - y_0)\sum y_{t-1}$

$$= T^{-2}(\beta T + u_T - u_0)\sum(\beta t - \beta + u_{t-1})$$

$$= T^{-1}\beta^2\sum t - \beta^2 + \beta T^{-1}\sum u_{t-1} + T^{-2}\beta(u_T - u_0)\sum t$$

$$- \beta T^{-1}(u_T - u_0) + T^{-2}(u_T - u_0)\sum u_{t-1}.$$

Subtracting (iv) from (iii), we obtain after some simplifications:

$$T^{-1}\sum y_{t-1}(y_t - y_{t-1}) - T^{-2}(y_T - y_0)\sum y_{t-1}$$

$$= \frac{\beta}{2}(u_T + u_0) - \frac{\beta}{2}T^{-1}(u_T - u_0) - \beta T^{-1}\sum u_{t-1}$$

$$- T^{-2}(u_T - u_0)\sum u_{t-1} + T^{-1}\sum u_t u_{t-1} - T^{-1}\sum u_{t-1}^2$$

$$\xrightarrow{\mathrm{D}} \beta X - \sigma_u^2 + \rho_1,$$

where $\xrightarrow{\mathrm{D}}$ denotes 'asymptotically distributed as' and X is as defined in Lemma A.1. Therefore,

$$T^2(\alpha^* - 1) \xrightarrow{\mathrm{D}} 12X/\beta - 12\left(\sigma_u^2 - \rho_1\right)/\beta.$$

(c) To prove part (b), we first need to prove the following lemma.

Lemma A.2. *If* $\{y_t\}_0^\infty$ *is a stationary series around a linear trend, then as* $T \uparrow \infty$:

$$S^{*2} \underset{p}{\rightarrow} \beta^2 + 2(\sigma_u^2 - \rho_1).$$

Proof.

$$S^{*2} = T^{-1}\sum(y_t - \alpha^* y_{t-1})^2$$

$$= T^{-1}\sum((y_t - y_{t-1}) - (\alpha^* - 1)y_{t-1})^2$$

$$= T^{-1}\sum\left[(y_t - y_{t-1})^2 - 2(\alpha^* - 1)y_{t-1}(y_t - y_{t-1}) + (\alpha^* - 1)^2 y_{t-1}^2\right],$$

(i) $\quad T^{-1}\sum(y_t - y_{t-1})^2$

$$= T^{-1}\sum(\beta + u_t - u_{t-1})^2$$

$$= T^{-1}\sum\left(\beta^2 + u_t^2 + u_{t-1}^2 + 2\beta u_t - 2\beta u_{t-1} - 2u_t u_{t-1}\right)$$

$$\underset{p}{\rightarrow} \beta^2 + 2(\sigma_u^2 - \rho_1),$$

(ii) $\quad 2T^{-1}(\alpha^* - 1)\sum y_{t-1}(y_t - y_{t-1})$

$$= T^{-1}\left[2T^2(\alpha^* - 1)T^{-2}\sum y_{t-1}(y_t - y_{t-1})\right]$$

$$\underset{p}{\rightarrow} 0,$$

(iii) $\quad T^{-1}(\alpha^* - 1)^2\sum y_{t-1}^2 = T^{-2}\left[T^4(\alpha^* - 1)^2 T^{-3}\sum y_{t-1}^2\right] \rightarrow 0,$

using part (a). The result follows. Finally,

$$t_{\alpha^*} = (\alpha^* - 1)\left[\sum(y_{t-1} - \bar{Y}_{-1})^2\right]^{1/2}/S^*,$$

$$T^{1/2}t_{\alpha^*} = T^2(\alpha^* - 1)\left[T^{-3}\sum(y_{t-1} - \bar{Y}_{-1})^2\right]^{1/2}/S^*$$

$$\overset{D}{\rightarrow} \sqrt{12}\,(\beta X - \sigma_u^2 + \rho_1)/\beta\left(\beta^2 + 2(\sigma_u^2 - \rho_1)\right)^{1/2}.$$

Appendix 2

The data set used in section 3 was provided by Robert J. Shiller. It is the same data set that he used in his paper with Sanford Grossman (1981) and his (1984) Brooking's paper. The description of the data is as follows:

(a) Dividend (1871–1984)	: Dividend per share adjusted to index, Standard and Poor composite, four-quarter total.
(b) PPI (1871–1984)	: January producers price index (formerly WPI). Data before 1900 is annual average.
(c) SP500 JAN (1871–1984)	: Nominal Standard and Poor composite stock price index for January.
(d) SP500 JUL (1871–1983)	: Nominal Standard and Poor composite stock price index for July.
(e) Earnings (1871–1984)	: Earnings per share adjusted to index, four-quarter total, Standard and Poor Composite [starting at 1926 from Standard and Poor Statistical Service; before 1926, Cowles series time 0,12338 (ratio of linked series in 1926)].
(f) Population (1889–1979)	: Population of the United States. After 1930 armed forces overseas included, before 1930 excluded. From the Department of Commerce, *Long Term Economic Growth 1865–1970* and *Current Population Reports*.
(g) CNT (1889–1979)	: Nominal consumption of nondurables and services. Starting from 1929 the U.S. National Income and Product Accounts (personal consumption expenditure total minus personal consumption expenditure on durable goods). Before 1929 total consumption expenditure [from Kendrick (1961)] minus durable component of flow of goods to consumers [from Kuznets (1961)] times 0,9764 (ratio of the series in 1929).
(h) CRT (1889–1979)	: Real consumption on nondurables and services. Same sources as CNT.
(i) PN (1889–1979)	: Standard and Poor Composite stock price annual average 1941–1943 = 10.

The constructed series are defined as follows:

(1)	PC (1889–1979)	= CNT/CRT
(2)	Per capita real (nominal) consumption (1889–1979)	= CRT/POP (CNT/POP)
(3)	Real dividend (PPI) (1871–1984)	= Dividend/PPI
(4)	Real earnings (PPI) (1871–1984)	= Earnings/PPI
(5)	Real dividend (PC) (1889–1979)	= Dividend/PC
(6)	Real earnings (PC) (1889–1979)	= Earnings/PC
(7)	Dividend/Price ratio (1871–1983)	= Dividend/ SP500JUL
(8)	Earnings/Price ratio (1871–1983)	= Earnings/ SP500JUL
(9)	Price (JUL) + Dividend (1871–1983)	= SP500JUL + Dividend
(10)	Price (JUL) + Dividend (Real) (1871–1983)	= (SP500JUL + Dividend)/PPI
(11)	Real SP500JUL (1871–1983)	= SP500JUL/PPI
(12)	Real SP500JAN (1871–1984)	= SP500JAN/PPI
(13)	Price index + Dividend (1889–1979)	= PN + Dividend
(14)	Price index + Dividend (Real, PC) (1889–1979)	= (PN + Dividend)/PC
(15)	Price index + Dividend (Real, PPI) (1889–1979)	= (PN + Dividend)/PPI
(16)	Real SP500JAN (PC) (1889–1979)	= SP500JAN/PC
(17)	Real SP500JUL (PC) (1889–1979)	= SP500JUL/PC

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